

DL & GenAI Project — Smart MCQ Solver

Milestone 5: Ensembling and Inference Strategies

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Overview

Milestone 5 focuses on improving prediction quality beyond what any single model can achieve by combining multiple fine-tuned transformer models through ensemble techniques. Rather than relying on the output of one model alone, ensembling aggregates probability distributions from several models, reducing individual model errors and producing more calibrated and robust predictions.

In this milestone, two transformer models — `microsoft/deberta-v3-small` and `roberta-base` — are independently fine-tuned on the MCQ training set and their softmax probability outputs are combined using simple average and weighted ensemble strategies. Additionally, Test Time Augmentation (TTA) is explored as a complementary inference-time technique. The final weighted ensemble submission is generated for all 500 test rows and evaluated on the first 100 validation samples using `mAP@3`.

Models and Training Setup

Input Formatting

Both models receive the same structured input text built from each training row:

Question: {prompt}

A) {option_A}

B) {option_B}

C) {option_C}

D) {option_D}

E) {option_E}

This flat format avoids the multiple-choice head used in Milestone 2 (DistilBERT) and instead frames MCQ answering as a 5-class sequence classification task, which is more flexible and works with any standard `AutoModelForSequenceClassification`.

Training Configuration

Both models are trained with identical hyperparameters for a fair comparison:

Hyperparameter	Value
Max sequence length	384 tokens
Batch size	8
Learning rate	2e-5
Epochs	3
Optimizer	AdamW ($\epsilon=1e-6$, weight decay=0.01)
Scheduler	Cosine with 10% warmup steps
Precision	bfloat16 (if supported), else float32
Gradient clipping	1.0
Train / Val split	80 / 20 (random_state=42, stratified)

A cosine schedule with warmup is used rather than a fixed learning rate — this prevents the optimizer from taking large steps early in training when the model weights are far from their optimal values, and gradually reduces the learning rate in later epochs.

NaN and Inf losses are explicitly skipped during training to prevent instability, which can occur with mixed precision on certain GPU configurations.

Ensemble Strategies

After both models are trained, inference on any input produces a 5-dimensional softmax probability vector — one probability per answer option. Ensembling combines these vectors across models before taking the final prediction.

Simple Average Ensemble

The probability vectors from DeBERTa and RoBERTa are averaged with equal weight:

$$P_{avg} = \frac{P_{DeBERTa} + P_{RoBERTa}}{2}$$

This gives both models equal influence in the final decision.

Weighted Ensemble

A weighted combination is used with DeBERTa contributing more heavily, reflecting its stronger architecture for language understanding tasks:

$$P_{weighted} = 0.7 \times P_{DeBERTa} + 0.3 \times P_{RoBERTa}$$

The weighted ensemble is used for the final Kaggle submission.

Test Time Augmentation (TTA)

TTA is applied to the first 50 test rows using DeBERTa only. Two versions of each input are created — the original structured prompt and an instruction-augmented version that prepends: “Answer the following multiple-choice question carefully:” The probability outputs from both passes are averaged to produce the final TTA prediction.

Results and Evaluation

Single-Row Inference (Test Row 25)

Single-row inference was run on test row 25 to verify model outputs and ensemble behaviour before running on the full test set. DeBERTa predicted option **C** with a confidence of 0.2468 (Q1). The simple average ensemble predicted option **E** (Q2), as did the weighted ensemble (Q3). The final top-3 prediction string for row 25 was **E C B** (Q4), which becomes the submission format for that row.

Ensemble Impact on First 100 Test Rows

Metric	Value
Rows where Weighted Ensemble Top-1 differs from DeBERTa alone (Q7)	74 / 100
Rows with positive confidence gain after ensembling (Q8)	100 / 100
Rows with at least one change in Top-3 ordering (Q9)	74 / 100

The ensemble changed the Top-1 prediction for 74 out of 100 rows compared to DeBERTa alone, and every single row showed a positive confidence gain after ensembling. This confirms that RoBERTa’s signal is meaningfully complementary rather than redundant. Q9 matching Q7 exactly indicates that whenever the Top-1 changed, the Top-3 ordering also changed — there were no cases where only lower-ranked options shifted.

TTA Results (Q6)

TTA produced zero Top-1 prediction changes across the 50 test rows evaluated. This is expected — DeBERTa has already been fine-tuned on structured MCQ inputs, so adding an instruction prefix causes minimal change in its attention patterns. TTA is more impactful on zero-shot or lightly fine-tuned models.

mAP@3 on First 100 Validation Rows

Configuration	mAP@3
Weighted ensemble (DeBERTa 3ep + RoBERTa 3ep)	1.0000
Weighted ensemble (DeBERTa 3ep + RoBERTa 1ep)	0.7767

The 3-epoch weighted ensemble achieves a perfect mAP@3 of 1.0000 on the first 100 validation rows. This aligns with the near-duplicate structure of the dataset established in the EDA — the training and validation sets share highly similar prompts, allowing a well-converged model to achieve near-perfect local scores. The 1-epoch RoBERTa variant drops performance significantly to 0.7767, demonstrating that an under-converged model hurts the ensemble rather than adding useful diversity.

Final Submission

The weighted ensemble is run on all 500 test rows using batched inference with batch size 16. The top-3 predictions per row are saved to `outputs/submission.csv` in the required format: space-separated answer letters ranked by confidence (e.g. E C B).

Observations and Insights

Flat classification vs. multiple-choice head. Unlike Milestone 2 which used `AutoModelForMultipleChoice` with a dedicated multiple-choice head, this milestone frames the task as 5-class sequence classification. This approach is simpler to implement, avoids the need for custom data collators, and scales naturally to any number of options without architectural changes.

DeBERTa is the stronger base model. The 0.7/0.3 weighting towards DeBERTa reflects its architecture advantage — DeBERTa uses disentangled attention over content and position, making it more effective at capturing subtle differences between answer options that differ in only a few words.

Ensembling adds diversity. The fact that 74 out of 100 rows had their Top-1 prediction changed by adding RoBERTa shows the two models have meaningfully different decision boundaries despite being trained on the same data. Every row showed positive confidence gain, confirming the ensemble is consistently more certain than either model alone.

TTA has no impact on fine-tuned models. Zero Top-1 flips across 50 rows confirms that instruction-prefix TTA is not a useful technique for a model already adapted to the task format. The model is robust to minor prompt rephrasing after fine-tuning.

Epoch count is critical for ensemble contribution. Replacing the 3-epoch RoBERTa with a 1-epoch variant dropped mAP@3 from 1.0000 to 0.7767 — a collapse of 0.2233 points. This shows that an under-trained model introduces noise rather than complementary signal, and that convergence is a prerequisite for a model to contribute positively to an ensemble.

Conclusion

Milestone 5 demonstrated a complete ensembling pipeline combining two independently fine-tuned transformer models — DeBERTa and RoBERTa — through weighted probability averaging, with Test Time Augmentation explored as an additional inference-time technique. The 3-epoch weighted ensemble achieved a perfect mAP@3 of 1.0000 on the local validation sample, and the 1-epoch experiment confirmed that model convergence is essential for positive ensemble contribution. The milestone establishes the final submission strategy for the project and delivers a production-ready submission covering all 500 test rows.